

HEALTHCARE AI + DIGITAL TWIN CHALLENGE

GlucoTwin

Predict the next glucose spike before it becomes the next clinical problem.

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THE PROBLEM

Metabolic care is still too often retrospective

Patients and clinicians lack a clear way to test how meals, sleep, and movement change near-term risk.

The gap

Post-meal glucose surges can form between routine check-ins. Static EHR snapshots and isolated readings miss continuous physiological context.

2-hour early warning

Glucotwin reframes the workflow around a 2-hour early warning.

One risk question drives the entire system

Will glucose rise by at least 40 mg/dL within the next 2 hours?

01

Observed window

Historical sensor data from $t-2h$ to t plus static EHR context

02

Prediction horizon

$t+15m$ to $t+2h$

03

Target

Spike = 1 when the maximum future rise reaches the threshold

04

Leakage guard

Future readings are removed before features are built or models are trained

THE DIGITAL TWIN

The twin turns disconnected signals into a living patient state

01

Ingest

EHR profile, CGM, heart rate, HRV, steps, sleep, and meal carbohydrates

02

Remember

Maintain a rolling 15-minute sensor history

03

Compute

Update 35+ trends, rolling statistics, meal lags, and physiological interactions

04

Act

Predict, explain, and simulate the next intervention

The technical stack is modular from data to dashboard



Data & modeling

Python 3.10+, synthetic physiology generator, preprocessing, StandardScaler



AI / ML

Logistic Regression, Random Forest, XGBoost, SHAP TreeExplainer



Backend

FastAPI async REST endpoints for patients, prediction, explanation, simulation, and benchmarks



Frontend

React 18, Vite, Tailwind CSS, Lucide icons, Recharts

The architecture closes the loop in five layers

1	2	3	4	5
Data ingestion EHR + 15-minute wearables + nutrition	Feature engine Trends, rolling statistics, meal lag, sleep and stress indices	Digital Twin core State history, recalibration, and state cloning	ML + XAI XGBoost risk probability and SHAP feature attributions	Clinician UI FastAPI, React, signal charts, alerts, and scenario sliders

XGBoost catches the risky moments

Optimized for high recall, the benchmark flags most future glucose spikes while keeping the signal measurable for clinical review.

Metric	Score	Readout
Precision	0.2831	Prioritizes coverage over alert volume
Recall	0.8782	Catches 87.82% of positive events
F1	0.4272	Balances precision and recall
ROC-AUC	0.8716	Strong ranking across thresholds
PR-AUC	0.5053	Useful signal under class imbalance

The benchmark is a research signal—not a diagnosis or treatment recommendation.

Every prediction comes with a why

SHAP turns the XGBoost score into a transparent set of risk drivers and protective signals.

Risk drivers

- Current glucose
- Meal carbohydrates
- Rising glucose trend
- Sleep deficit

Protective signals

- Recent steps
- HRV recovery
- Stable recent trajectory
- Recovery after movement

A ranked explanation helps a researcher inspect the model before acting on its output.

Test an intervention before the next spike

What-if sliders let a user explore how a small behavior change could shift the predicted trajectory.

Before

- Large meal load
- No post-meal movement
- Low recovery signal
- Higher predicted risk



simulate

After

- Adjusted carbohydrate input
- Short walk added
- Recovery restored
- Lower predicted risk

The simulation is a comparison tool: change inputs, inspect the response, and keep the original context visible.

From model output to a usable research workflow

Built for iteration

- FastAPI service for model and DigitalTwin endpoints
- React 18 interface bundled with Vite
- Recharts for glucose trajectories and model signals

One clear loop

1. Review the current glucose context
2. Inspect SHAP drivers and protective signals
3. Move what-if sliders for meals, movement, sleep, and recovery
4. Compare the before/after trajectory

A focused dashboard keeps the prediction, explanation, and experiment in one place.

A foundation for safer, more explainable glucose research

What is in the prototype

- Synthetic data pipeline
- 35+ engineered features
- Stateful DigitalTwin class
- XGBoost benchmark with SHAP explainability

Next on the roadmap

- Validate against richer cohorts
- Calibrate thresholds with domain experts
- Test longitudinal intervention scenarios
- Harden monitoring, auditability, and consent flows



Research only

This prototype is not medical advice, is not a diagnostic device, and must not be used to make treatment decisions. Any future clinical use requires rigorous validation, oversight, and regulatory review.